

E-Commerce Customer Segmentation & Retention Analysis

Customer Analytics Project

1. Introduction

This project analyzes e-commerce customer purchasing behavior to identify customer segments and understand customer retention and churn patterns. RFM analysis is used to classify customers according to their purchasing behavior.

2. Objectives

- Clean and prepare the e-commerce dataset.
- Analyze customer purchasing behavior.
- Perform RFM analysis.
- Segment customers based on RFM scores.
- Analyze purchase trends and customer churn.
- Provide actionable business recommendations.

3. Dataset Description

The dataset contains customer and transaction information including Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Returns, Age, Gender and Churn.

4. Data Cleaning

The Purchase Date column was converted into datetime format. Missing values and duplicate records were checked, duplicate records were removed, and the data types of the columns were verified. A Month column was also created for monthly purchase analysis.

5. RFM Analysis

Recency: Number of days since the customer last purchased.

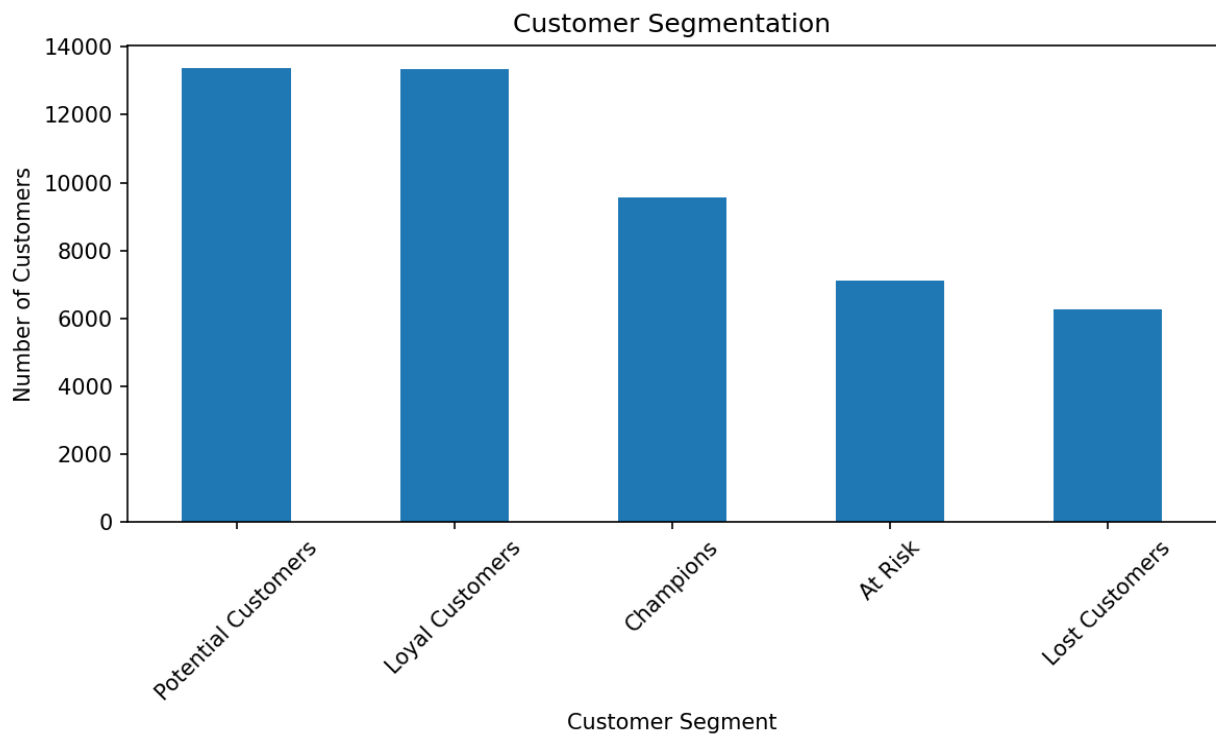
Frequency: Number of purchases made by the customer.

Monetary: Total amount spent by the customer.

RFM scores ranging from 1 to 5 were calculated for each customer.

6. Customer Segmentation

Customers were classified into five groups based on their RFM scores: Champions, Loyal Customers, Potential Customers, At Risk, and Lost Customers.

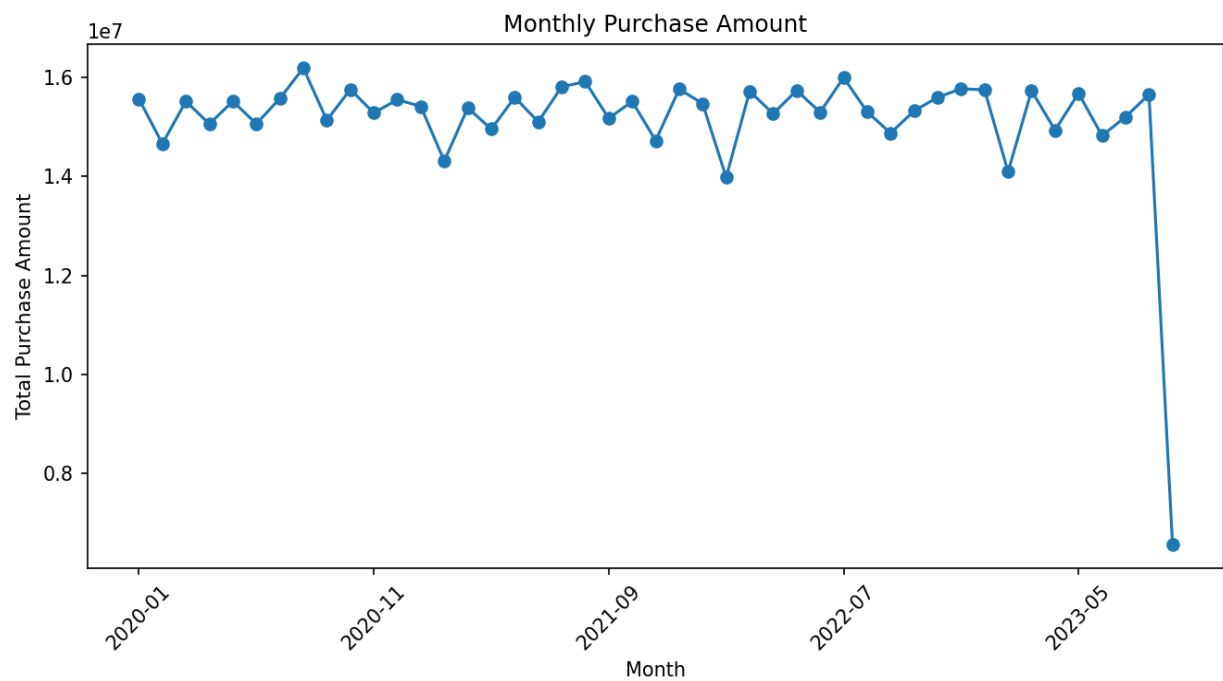


7. Segment Analysis

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary
At Risk	7110.0	370.70	3.26	8307.15
Champions	9573.0	76.73	7.90	22509.13
Lost Customers	6282.0	654.15	2.18	5338.36
Loyal Customers	13329.0	175.54	6.00	16574.61
Potential Customers	13367.0	240.03	4.31	11397.02

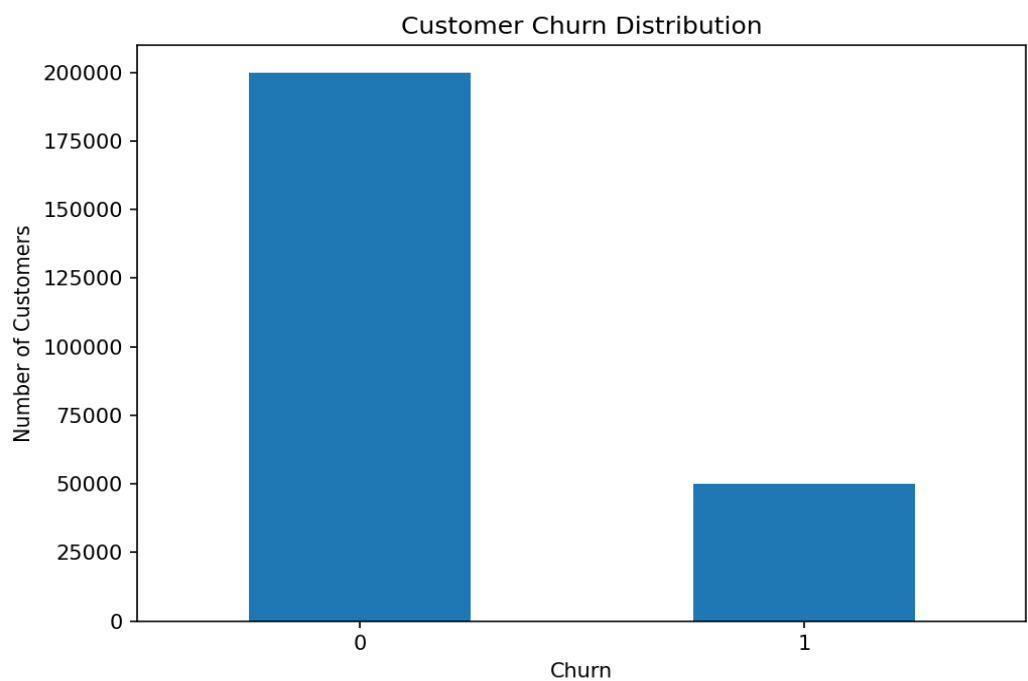
8. Purchase Trend Analysis

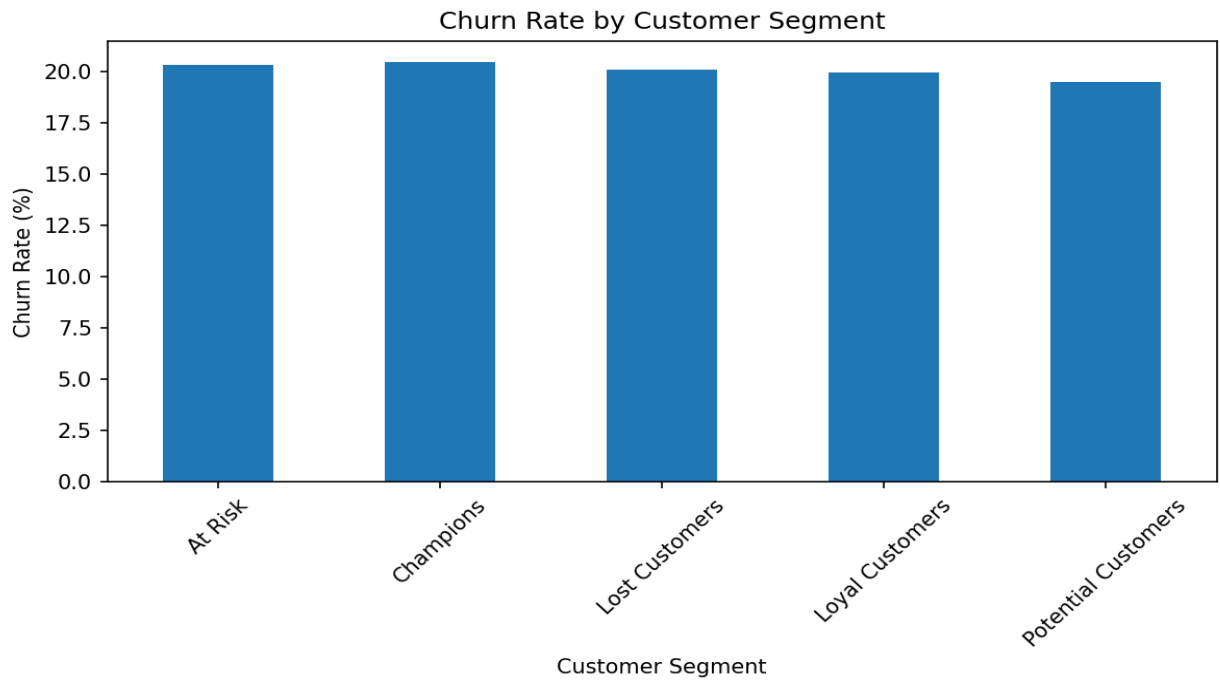
Monthly purchase amounts were analyzed to identify changes in customer purchasing activity over time.



9. Churn Analysis

The calculated overall churn rate is **20.05%**. Churn rates were also compared across customer segments to identify groups requiring retention efforts.





10. Actionable Recommendations

- **Retain Champions:** Provide loyalty rewards, exclusive offers, and early access to new products.
- **Engage Loyal Customers:** Use personalized recommendations and targeted promotions to increase purchase frequency.
- **Reactivate At-Risk Customers:** Send win-back campaigns, discounts, and personalized offers.
- **Convert Potential Customers:** Encourage repeat purchases through personalized recommendations and limited-time offers.
- **Reduce Customer Churn:** Identify customers showing declining engagement and launch targeted retention campaigns.

11. Conclusion

The analysis demonstrates how e-commerce transaction data can be used to understand customer purchasing behavior. RFM segmentation, purchase trend analysis, and churn analysis provide useful insights for developing targeted marketing and customer retention strategies.